

GroupTrack V2.5 — Feature Schedule and Architecture Plan

Merged from V25_BACKLOG.md + V2.4 session deferred items + May 8, 2026 new items
Audience: future-Fred and any AI assistant continuing the work

Strategic Context

V2.5 bridges 'convoy app for radio owners' to 'free GPS tracker for any rider.' Two goals: (1) standalone Gaia/onX parity, (2) spatial database architecture for national trails. Map Viewer = planning/exploring hub. Convoy Map = operational hub.

Master Schedule — Dependency-Ordered

Seq	Item	Pri	Est.	Depends	Source
0	DEV_ENVIRONMENT v4 + recommit fix	P1	1d	None	V2.4 session
1	navigationBarsPadding sweep	P1	1d	None	V2.4 session
2	Spatial DB: AWS (PostGIS/MySQL spatial)	P1	3-5d	None	May 8 new
3	Spatial DB: Android (Spatialite/Room)	P1	3-5d	Seq 2	May 8 new
4	Waypoints: trailheads + POIs	P1	3-5d	Seq 2,3	May 8 new
5	Cart selection: radio list panel	P1	2-3d	None	May 8 new
6	National trail: GeoPackage + 3 engines	P1	2-3w	Seq 2,3	Backlog
7	Trail source catalog (BLM/USFS/state)	P1	2-3d	Seq 6	Backlog
8	Mileage / trip odometer	P1	2-3d	None	Backlog
9	Multi-select + bulk actions	P2	2-3d	None	Backlog #2
10	Track naming (location-based)	P2	2d	None	Backlog #3
11	Native GPS speed (getSpeed())	P2	1d	None	Backlog #10
12	Track import to Map Viewer hub	P2	1d	None	V2.4 session
13	SubMenu nav root cause	P2	1d	None	V2.4 session
14	Higher-res GPS (1s option)	P2	1d	None	Backlog
15	Track stats (dist/elev/time/speed)	P2	2-3d	Seq 8	Backlog
16	Speed warmup tightening	P2	1d	Seq 11	Backlog
17	Solo rider onboarding	P2	2-3d	None	Backlog
18	Orphan track cleanup button	P2	1d	None	Backlog #8
19	Esri migration planning	P2	2-3d	None	Backlog #4
20	Cache versioning + Refresh Tiles	P3	2-3d	None	Backlog #6
21	Tile interceptor logging	P3	1d	None	Backlog #7
22	Auto-clear launcher cache	P3	0.5d	None	Backlog #9

Phase 0 — Foundation (Seq 0-1) — Before Any Feature Code

Seq 0: DEV_ENVIRONMENT v4 + Recommit Script

- Update v3 to v4: add release build commands (assembleGoogleRelease, bundleGoogleRelease), AAB process, google-services.json debug client requirement.
- Recommit script: v6 works, v7/v8 broken. Recover v6 from git (dbf7e2549), diff v7/v8, fix or rewrite.
- Run fixed recommit to update all docs before starting V2.5 code.

Seq 1: navigationBarsPadding Sweep

V2.4 applied to Work With Tracks + Import Screen. Audit every full-screen composable and add where bottom content is cut off. Also fix hiltViewModel deprecation (11 warnings) and Divider to HorizontalDivider rename.

Phase 1 — Architecture (Seq 2-7) — Spatial DB + Trails + Cart Selection

Seq 2-4: Spatial Database + Waypoints

Seq 2 — AWS Server (3-5 days)

- Evaluate PostGIS (PostgreSQL) vs MySQL spatial on existing RDS.
- Waypoints table exists (UUID PK, coords + name, FK from rides). Add geometry column, spatial index, SRID 4326 (WGS84).
- API: CRUD waypoints, spatial queries (within radius, within bounding box).

Seq 3 — Android Local (3-5 days)

- SpatialLite or Room with R-tree indices. Offline-first, sync to server when connected.
- Replaces file-based storage. Eventually replaces GeoJSON trail loading.
- Query patterns: nearest trailhead, trails in viewport, points within radius.

Seq 4 — Waypoints UI (3-5 days)

- Trailhead waypoints: mark and name on map. POI waypoints along rides: fuel, gates, hazards, scenic.
- Display on both Convoy Map and Map Viewer. Sync device to server. Share between riders.

Seq 5: Cart Selection Redesign

- Remove convoy long-press for cart selection entirely.
- Add RIDERS button to convoy map HUD (alongside TRAILS).
- Panel shows all connected radios: callsign, position age, signal strength, battery.
- Tap a radio to center map on that cart. Lead cart indicator (star/highlight).

Seq 6-7: National Trail Model

Seq 6 — GeoPackage Architecture (2-3 weeks)

- grouptrack.gpkg replaces 22 MB Utah GeoJSON. Three engines: ingestion, distribution, local display.
- Ingestion: UGRC, BLM, USFS per region. Distribution: region downloads paired with tile sets.
- Phase 1: Utah pilot (5,732 trails). Phase 2: AZ, NV, ID, CO. Phase 3: National.

Seq 7 — Trail Source Catalog (2-3 days)

- Catalog state/federal GIS portals. Each state publishes differently.
- Known gaps: Parowan Gap (BLM GTLF), Arizona Strip (AZ BLM GeoJSON).
- Define user workflow for selecting and downloading regional trail files.

Phase 2 — Features (Seq 8-18)

Seq 8: Mileage / Trip Odometer (P1)

Running total on HUD during ride. Summary in post-ride stats. GPS track point distance accumulation.

Seq 9: Multi-Select + Bulk Actions (P2)

From V25_BACKLOG.md #2 — detailed design agreed:

- Long-press track to enter multi-select. Checkboxes on all rows.
- Select All / Deselect All. Bottom toolbar: Delete, Share, Move to Downloads, Fix Creation Date.
- Rename excluded (unique input needed). Open on Map excluded (single file only).
- ConvoyTrackOps additions: deleteTracksBatch, shareTracksBatch, copyTracksToDownloadsBatch.

Seq 10: Track Naming — Location-Based (P2)

From V25_BACKLOG.md #3:

- Use Geocoder on initial GPS fix to get location name.
- Suggest filename: nearest place + date. User confirms or edits.
- Affects ConvoyGpsService.kt recording filename and ConvoyConfig.kt naming pattern.

Seq 11: Native GPS Speed (P2)

From V25_BACKLOG.md #10: Drop 60-second windowed calculation, use Location.getSpeed(). Faster updates, eliminates 0 mph warmup. Affects ConvoyGpsService.kt.

Seq 12: Track Import Move to Map Viewer (P2)

- V2.4 shipped import via Work With Tracks button + Map Viewer TRACKS dropdown.
- V2.5: Map Viewer becomes the single hub. Move Work With Tracks into Map Viewer.
- Remove Work With Tracks from SubMenu MAP SETTINGS (replaced by Map Viewer).

Seq 13: SubMenu Navigation Root Cause (P2)

V2.4 diagnosis: ConvoyScreen ModalBottomSheet cannot navigate to ConvoyTrackImport from Convoy start destination. Swap test proved callback fires. Other routes work. LaunchedEffect failed. Workaround in place. Root cause: likely Compose Navigation issue with new routes.

Seq 14-18: Standalone Polish

- **14 — Higher-res GPS (1d):** 1-second recording option (currently 3s).
- **15 — Track stats (2-3d):** distance, elevation gain, moving time, max speed. Depends on Seq 8.
- **16 — Speed warmup fix (1d):** faster initial display. Depends on Seq 11.
- **17 — Solo onboarding (2-3d):** first-launch tutorial without radio assumption. Marketing copy.
- **18 — Orphan cleanup (1d):** 'Delete All In-Progress' button with confirm + count + size. Never auto-delete.

Phase 3 — Map Infrastructure (Seq 19-22)

Seq 19: Esri Migration Planning (P2)

From V25_BACKLOG.md #4-5 — investigation completed:

Approach	Cost	Effort	Off-road Coverage
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Stay Esri legacy	\$0	None	Degrading slowly
Migrate Esri ibasemaps	\$0 free tier, then paid	Low (URL + token)	Same as legacy
Google Maps SDK native	\$0 unlimited	High (WebView rewrite)	Excellent
OpenStreetMap	\$0	Medium	Poor off-road

Decision deferred: stay on legacy until forced. Cached tiles work indefinitely. If rewriting anyway, Google Maps SDK gives unlimited free + better performance.

Seq 20-22: Tile Improvements (P3)

- **20 — Cache versioning (2-3d):** metadata per region (source URL, timestamp, version). 'Refresh Region' action. Solves double-label issue from V2.4.
- **21 — Tile interceptor logging (1d):** add Log.d to all intercepted URLs, not just convoy://.
- **22 — Auto-clear launcher cache (0.5d):** fix duplicate icon. Documentation or post-install hook.

V2.4 Track Manager Architecture (Built — Context for V2.5)

From V25_BACKLOG.md — three-file split for reusability:

ConvoyTrackOps.kt — pure file logic, no UI, no Compose. Object singleton.

- tracksDir(), downloadsDir(), listTracks(), isInProgress(file)
- renameTrack, deleteTrack, copyToDownloads, shareTrack, formatSize
- extractEarliestTime (GPX/KML), importTrackFile (single/multi split), fixDateFromContent, fixDatesForFiles

ConvoyTrackDialogs.kt — reusable Composable dialogs.

TrackAlertDialog, RenameTrackDialog, DeleteTrackDialog

ConvoyTrackExportSheet.kt — 'Work With Tracks' full-screen manager.

Name preserved for caller compatibility. Sort (4-state cycle), filter tabs (All/Saved/In-Progress), search, per-track actions, bulk fix dates, Import button.

ConvoyTrackImportScreen.kt — in-app file browser (NEW in V2.4).

Scans Downloads for GPX/KML. Multi-select, progress dialog, recap dialog. Date correction on import. Accessible from Work With Tracks header + Map Viewer TRACKS dropdown.

Standing Architecture Rules

- **Reusability:** track features go in ConvoyTrackOps (logic) and ConvoyTrackDialogs (UI), not in calling screens.
- **No auto-delete:** NEVER programmatically delete user files. All deletes require explicit user action + confirmation.
- **Date preservation:** earliest GPS time from content = file mtime. Set on import via setLastModified.
- **Function naming:** preserve public function names when refactoring. ConvoyTrackExportSheet kept old name for callers.
- **File organization:** convoy/ConvoyTrack*.kt for track features. Pure logic separate from Compose UI.
- **Build discipline:** batch changes, verify via grep, one build per problem. 20 min builds = no wasted cycles.
- **Never commit V3_FEATURES_ENABLED = true.** Always restore to false after V3.0 dev build.

Known Technical Debt

Item	Details	Fix In
google-services.json	SHA-256 commit overwrote debug client. Restored May 8. Future releases	Seq 0
ModalBottomSheet + Nav	SubMenu can't navigate to new routes from start destination. Workaround	Seq 1
hiltViewModel deprecation	11 warnings. Migrate to androidx.hilt.lifecycle.viewmodel.compose.	Seq 1
Divider deprecation	ConvoyTrackExportSheet. Rename to HorizontalDivider.	Seq 1
ServiceKeepAliveWorker	Missing foregroundServiceType. Pre-existing from Meshtastic.	Backlog
PAYWALL_ENABLED	False for testers. Re-evaluate for V3.0.	V3.0
Recommit script	v6 works, v7/v8 broken.	Seq 0
WebView cache	Internal HTTP cache can differ from interceptor. Causes online/offline	Seq 20

V3.0 Items (Awareness Only)

- **Package migration:** com.geeksville.mesh to com.grouptack.android. V3.0 pre-release gate. Enables side-by-side install.
- **Cloud sync:** tracks/routes/waypoints to web account. Premium. Depends on spatial DB.
- **Tile cloud sync:** upload tiles to S3, download to any device. Delta sync. Like iCloud Photos for maps. Org service revenue.

- **Map Manager Phase C:** auto-download on ride enrollment, PROTECTED/PURGEABLE states, 2GB cap, storage dashboard.
- **Subscription:** \$3.00/month Google Play + Apple IAP. Single tier.
- **Dashboard:** 9 P1 issues (DASH-01 to DASH-16). Downstream: Ride Detail, My Rides, Explore, Profile, Create Ride.